COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument is a recurring technical step in min-max constructions of embedded minimal hypersurfaces: a cover-refinement on the parameter interval, followed by a bookkeeping argument that turns local near-critical savings into a global supremum improvement. Each author re-derives it in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g), direct minimization over a fixed homology class may produce only degenerate configurations, and the relevant critical points are typically of saddle type. The resolution, following Birkhoff (1917) for closed geodesics and Almgren–Pitts for higher-dimensional varifolds, is a *min-max* construction; regularity of the output is Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set $\operatorname{Sing}\Sigma$ has Hausdorff dimension at most $n-7$.*

The core technical result of Pitts's proof, used in every subsequent treatment [CDL03, DLT13, DLR18, DL16, MN14], is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Theorem 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integral varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Theorem 1.2 reduces to a one-parameter claim on the sweepout [DLT13, §3.2]:

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ [DLT13, §3.2] then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

The scalar formulation of the proof of the Claim, which is the precise theorem we formalize, reads as follows.

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Theorem 1.3 (Extracted scalar version)**.** *Let X be a pseudo-metric space with an abstract paired-cover structure. Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$ and $m_0 > 0$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open neighborhood $U_t \ni t$, a paired cover c_t in X, and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exists $f' : [0,1] \to \mathbb{R}$ with*

$$\sup f' \;\leq\; m_0 - \frac{1}{4N}.$$

Translation from DLT's setting to Theorem 1.3: the boundary $\partial\Omega_t$ becomes a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; and the almost-minimizing hypothesis is no longer needed.

The Lean library actually proves a more general statement, of which Theorem 1.3 is a direct specialization: the parameter interval $[0,1]$ may be replaced by any compact, nonempty topological space K, and the specific pair $(1/N, 1/(4N))$ by arbitrary scalars (δ, ε) with $0 < \varepsilon \leq \delta$. The input is then a finite covering of the δ-super-level set of f carrying per-piece replacements with uniform saving ε.

Theorem 1.4 (Abstract core)**.** *Let K be a compact, nonempty topological space, $f : K \to \mathbb{R}$ a continuous energy with $m_0 = \sup f$, and let $0 < \varepsilon \leq \delta$. Suppose there is a finite family of pieces $\{p_l \subseteq K\}_{l \in \iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \geq \varepsilon\}$ with: (I) $f - E_l \geq s_l$ on p_l; (II) every $t \in K$ lies in at most two pieces; (III) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$. Then there exists $f' : K \to \mathbb{R}$ with*

$$\sup f' \;\leq\; m_0 - \varepsilon.$$

Theorem 1.3 follows by taking $K = [0,1]$, $\delta = 1/N$, $\varepsilon = 1/(4N)$, and running a 1D cover-refinement construction on the local witnesses to produce the family $\{p_l\}$ with two-fold overlap. Both theorems reappear in §2: Theorem 1.4 as the library's abstract API (Theorem 2.7, §2.2), and Theorem 1.3 as the one-parameter application (Theorem 2.8, §2.3).

Scope. This note formalizes the proof of the Claim (the combinatorial argument on $[0,1]$) as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships a single abstract theorem (§2.2) parametric in the sweepout space together with a one-parameter specialization to $[0,1]$ (§2.3); the formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space), `X` a pseudo-metric space (the ambient space whose covers witness replacements), and `f : K → ℝ` a continuous scalar energy.

In DLT's proof of Theorem 1.2, [DLT13, Lemma 3.1] supplies, for each near-critical parameter t, an $\mathcal{H}^n$-disjoint pair of open sets $(U'_{1,t}, U'_{2,t})$ in M^{n+1} inside which a local replacement is performed. We abstract this pair as an instance of a class, stripped of its GMT content.

Definition 2.1 (`PairableCover` class)**.** [✓ `PairableCover`]

A `PairableCover X` instance declares, for each abstract `c : Cover`, a left region and a right region in X and a scalar diameter, subject to two axioms: the two regions are disjoint, and overlapping combined regions of diameter-comparable covers are ordered by inclusion.

DLT's replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ at a near-critical t has its Hausdorff measure decreased by at least $1/(4N)$ inside U'_t. We keep only the scalar *replacement energy* $E : K \to \mathbb{R}$ and the uniform saving inequality; the geometric family is discarded.

Definition 2.2 (`LocalWitness` structure). [✓ `LocalWitness`]
A `LocalWitness K X f t` ε carries an open neighborhood $U \ni t$ in K, a cover `c : Cover`, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

DLT's proof works on the set $K \subset [0,1]$ of parameters at which the sweepout is within $1/N$ of the min-max value. We use the same convention.

Definition 2.3 (Near-critical set). [✓ `nearCritical`]
For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

At the end of DLT's construction the data is a refined family $\{\overline{J_l}\}$ covering K, with each J_l carrying a replacement whose energy saving is at least $1/(4N)$ and with two-fold overlap. We abstract this output: the near-critical threshold $1/N$ becomes a parameter δ and the saving floor $1/(4N)$ a parameter $\varepsilon \leq \delta$.

Definition 2.4 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]
A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

Before DLT's refinement, a Lebesgue-number argument on the compact set K produces a finite covering $\{I_i\}$ of open intervals, each centered at a $t_i \in K$. In our formalization the interval center c_i and the witness center w_i are recorded separately: grid points need not themselves lie in K.

Definition 2.5 (`InitialCover` structure). [✓ `InitialCover`]
Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries a finite family of open intervals $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ for $i = 1, \ldots, n$, each paired with a witness center $w_i \in$ NC and a local witness at w_i with saving $1/(4N)$, subject to (a) spacing: $c_i + r_i < c_{i+2} - r_{i+2}$ for all valid i; (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: NC $\subseteq \bigcup_i I_i$.[1]

DLT's refinement is described as an iteration: each I_j with $j \geq 2$ is either kept whole or split into two halves. The state of this iteration after ℓ moves is the following.

Definition 2.6 (`PartialRefinement` structure). [✓ `PartialRefinement`]
Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. **Public API: abstract core.**

Theorem 2.7 (Abstract core). [✓ `exists_sup_reduction_of_cover`]
Input. *A compact, nonempty topological space K; a continuous $f : K \to \mathbb{R}$; $m_0 = \sup f$; scalars $0 < \varepsilon \leq \delta$; a* `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ *(Definition 2.4).*
Output. *There exists $f' : K \to \mathbb{R}$ with $\sup f' \leq m_0 - \varepsilon$.*

[1] Conditions (a) and (b) transcribe [DLT13, §3.2]'s spacing and subset conditions.

2.3. Public API: one-parameter application.

Theorem 2.8 (One-parameter application). [✓ `exists_sup_reduction`]
Input. *A pseudo-metric space X with a* `PairableCover X` *instance (Definition 2.1); a continuous $f : \mathbf{I} \to \mathbb{R}$; $m_0 \in \mathbb{R}$ with $m_0 > 0$ and $m_0 = \sup f$; $N \in \mathbb{N}$ with $N > 0$; for every $t \in \mathrm{NC}$ a* `LocalWitness` *at t with saving $1/(4N)$ (Definition 2.2).*
Output. *There exists $f' : \mathbf{I} \to \mathbb{R}$ with $\sup f' \le m_0 - 1/(4N)$.*
Specializes Theorem 2.7 to $K = \mathbf{I}$, $\delta = 1/N$, $\varepsilon = 1/(4N)$.

2.4. Internal lemmas.

The declarations below are consumed by Theorem 2.8; proofs are sketched in §3.

Proposition 2.9 (Compactness of NC). [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]
If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- ✓ `exists_initialCover` : existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` : existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` : at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` : the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.
- ✓ `exists_refinement` : packages the 1D construction output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$; the input to Theorem 2.7.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.8 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. Theorem 2.7 closes.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover). [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.8, an initial cover (Definition 2.5) exists.

Construction. Compactness of NC (Proposition 2.9) plus the open cover $\{U_t\}_{t \in \mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \dots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case). [✓ `step_zero`]
step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

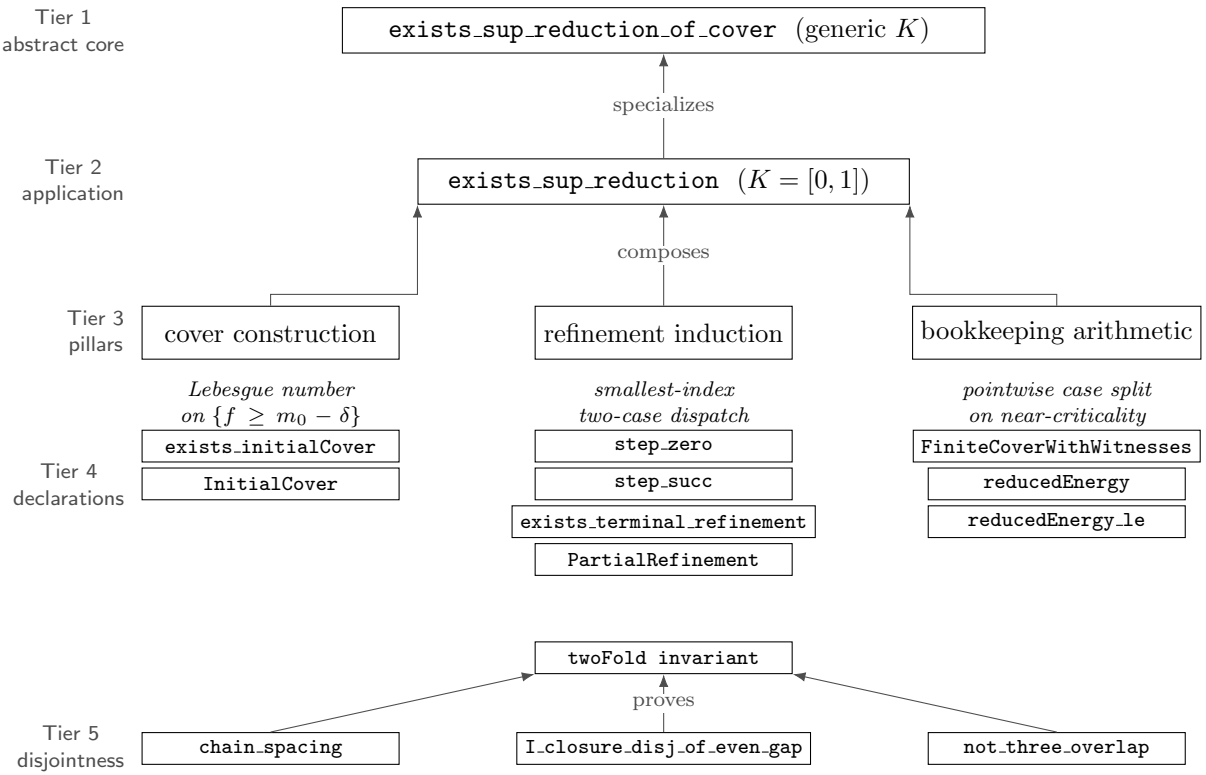

FIGURE 1. `CombArg` v0.1 architecture. The abstract core (Tier 1) is parametric in the sweepout space; the one-parameter application (Tier 2) instantiates $K = [0, 1]$ and composes three construction pillars (Tier 3). Each pillar's key declarations appear in Tier 4. The disjointness sub-module (Tier 5) supplies the two-fold overlap invariant consumed by bookkeeping.

Definition 3.3 (Inductive step). [✓ `step_succ_at`]

Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \ldots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

Proposition 3.4 (Termination). [✓ `exists_terminal_refinement`]

Starting from step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Argument. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr′ with $i^* \notin$ rem(pr′) and rem(pr′) $\subseteq$ rem(pr) $\setminus \{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell + 1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

[2] A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

3.3. **Stage C: assembly.** Stage C packages the refined covering as a `FiniteCoverWithWitnesses` (Definition 2.4), which the core theorem turns into the sup-reducing f'. For comparison, we quote the DLT passage in which the packaging and the final arithmetic live side by side.

> **DLT** [DLT13, §3.2, Step 2]**.** We apply Lemma 3.1 to conclude the existence of a family $\{\partial\Omega_{i,t}\}$ with: $\Omega_{i,t} = \Omega_t$ if $t \notin (a_i, b_i)$ and $\Omega_{i,t} \setminus U'_i = \Omega_t \setminus U'_i$ if $t \in (a_i, b_i)$; $\mathcal{H}^n(\partial\Omega_{i,t}) \leq \mathcal{H}^n(\partial\Omega_t) + 1/(4N)$ for every t; $\mathcal{H}^n(\partial\Omega_{i,t}) \leq \mathcal{H}^n(\partial\Omega_t) - 1/(2N)$ if $t \in (a_i + \delta, b_i - \delta)$. If $t \in (a_i, b_i) \cap (a_j, b_j)$ then $j = i + 1$ and $\mathrm{dist}(U'_i, U'_{i+1}) > 0$; define $\Omega'_t = \Omega_t$ outside $\bigcup J_i$, $\Omega'_t = \Omega_{i,t}$ inside a single J_i, and $\Omega'_t = [\Omega_t \setminus (U'_i \cup U'_{i+1})] \cup [\Omega_{i,t} \cap U'_i] \cup [\Omega_{i+1,t} \cap U'_{i+1}]$ on $J_i \cap J_{i+1}$. If $t \notin K$ then $\partial\Omega'_t$ loses at most $2 \cdot (4N)^{-1}$ in area; if $t \in K$ then $\partial\Omega'_t$ gains at least $(2N)^{-1}$ in some U'_i and loses at most $(4N)^{-1}$ in at most one U'_l, so $\mathcal{H}^n(\partial\Omega'_t) \leq \mathcal{H}^n(\partial\Omega_t) - (4N)^{-1}$.

Our formalization splits DLT's passage along the structural boundary between the cover and the arithmetic. The three bullets on $\{\partial\Omega_{i,t}\}$ are the data we bundle into a `LocalWitness` at each near-critical parameter: the scalar replacement energy $E_{\sigma(k)}$ plays the role of $\mathcal{H}^n(\partial\Omega_{i,t})$, the neighborhood U_t plays (a_i, b_i), and the $1/(4N)$ saving is the bullet "$\mathcal{H}^n(\partial\Omega_{i,t}) \leq \mathcal{H}^n(\partial\Omega_t) - 1/(2N)$" already at the cover-refinement stage (our Stage B produces pieces on which the witness saving holds). The observation "$t \in (a_i, b_i) \cap (a_j, b_j) \Rightarrow j = i + 1$" is our TwoFold invariant (Proposition 3.5). The case split at $t \notin K$ versus $t \in K$ is the two-branch case split in the pointwise argument for `reducedEnergy_le` inside the abstract core (§2.2). The piecewise definition of Ω'_t corresponds to the scalar formula $f'(t) = f(t) - \sum_{l:t\in\overline{J_l}} s_l$; DLT's piecewise construction is an explicit sum over the at-most-two pieces containing t. The $(4N)^{-1}$ appearing at the end of DLT's paragraph is our ε; the abstract core generalises the pair $(1/N, 1/(4N))$ to (δ, ε) with $\varepsilon \leq \delta$ and reads off the same arithmetic unchanged.

The present stage proves the two propositions that feed the core: the saving bound extends from open J_k to $\overline{J_k}$ (continuity), and the TwoFold invariant holds (from spacing (a)).

Proposition 3.5 (TwoFold via parity)**.** [✓ `terminal_twoFold`]

For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \ldots, \overline{J_L}$.*

Argument. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i - j| = 2(m + 1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \ldots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \ldots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound)**.** [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Argument. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.2 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Lemma 3.7 (Assembly)**.** [✓ `exists_refinement`]

A partial refinement pr *of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i yields a refinement of $(\mathbf{I}, X, f, m_0, N)$ in the sense of Definition 2.4, with $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and $E_k :=$ the replacement energy of the $\sigma(k)$-th witness.*

Argument. Invariant (I) is Proposition 3.6. Invariant (II) is Proposition 3.5. Invariant (III) holds by construction ($s_k = 1/(4N)$). Coverage of NC chains $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Theorem 2.8 now follows: chain Propositions 3.1 and 3.4, invoke Lemma 3.7, and apply Theorem 2.7.

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